

Mathematical modeling of the innate and adaptive immune systems in a viral infection

B. Chen-Charpentier^{b,1}

(b) Department of Mathematics, University of Texas at Arlington
Arlington, Texas 76019, USA.

1 Introduction

Viruses cause a large number of diseases in humans, animals and plants. In humans the viral diseases include the common cold (Rhinovirus/Enterovirus, Respiratory Adenoviruses), flu (Influenza), COVID-19 (SARS-CoV-2), measles, hepatitis, AIDS (Human Immunodeficiency Virus) and many others. The effects of a viral infection can be mild as for the flu to deadly as for COVID-19, AIDS and Ebola. Viruses are non-cellular entities. Each virus particle or virion consists of a nucleic acid core, an outer protein coating, and sometimes an outer envelope [9]. They replicate by penetrating a healthy cell, and using the cell's RNA or DNA to make copies of its genetic material, and then burst the cell producing many new virions. Vaccines and antivirals can be used to fight the infection. Vaccines, usually administered before the individual gets the disease, train the immune system to fight the virus. Antivirals are medications that reduce the effects and the duration of the disease. Rather than killing a virus directly, antivirals usually suppress the virus's ability to infect and multiply in the organism cells [28]. The human body has physical barriers such as the skin and cilia in the respiratory tract. But viruses can overcome these barriers. Then the innate immune system responds with inflammation, virus phagocytosis, and secretion of cytokines, chemical factors and proteins [9]. The innate immune system is non-specific and responds to the presence of any debris or pathogen. The adaptive immune system takes longer to act than the innate system but is more specific and has memory. Within-host mathematical models can help understand the very complex interactions between the different cells, chemical substances and virions. A basic model consists of three populations, susceptible cells, infected cells and virions. Their interactions and rates of change are given by ordinary differential equations [23, 25, 1]. An improvement is to add effector cells to kill the infected cells [22, 15] or to kill the virus [21, 18]. Both processes are added without giving explicit equations for the immune system in [13]. More elements of the immune system are included in [2, 4, 14]. There are other processes that have been observed for some virus within-host infections. One is the presence of cell to cell infection [11, 26]. Delays are also added in many papers including [27, 33, 34]. The effects of treatment are studied in [31, 3, 6, 32]. In this paper we introduce and analyze a new within-host virus infection model, based on ordinary differential equations. It includes healthy cells, infected cells, virions, and innate and adaptive immune system cells. It also includes cell to cell infection. Two more simple previously published models are included in the study for comparison purposes. Finally numerical simulations for a model of hepatitis B are done, first with no treatment and then with treatment.

¹bmchen@uta.edu

2 Methods

The first within-host virus infection model we study only includes three populations: susceptible cells, infected cells, and virus particles (virions) [23, 25, 1]. Susceptible cells are recruited and die naturally, and they become infected by invasion of a virus particle. The virus replicates inside the cell and then the cell bursts and releases a certain number of virions. These free virions can infect healthy cells or die. This death process is not specified but it is implied to be caused by the immune system. All parameters are non-negative. The equations that describe the model are:

$$\begin{aligned}\frac{dx}{dt} &= \lambda - \mu_x x - \beta x v \\ \frac{dy}{dt} &= \beta x v - \mu_y y \\ \frac{dv}{dt} &= B \mu_y y - \mu_v v.\end{aligned}\tag{1}$$

Here, x, y , and v , are the concentrations of susceptible cells, infected cells, and virions, respectively. λ is the recruitment rate of new susceptible cells, μ_x is their death rate, β is the infection rate, μ_y is the bursting rate of infected cells by the virus, B is the number of virions produced per infected cell, and μ_v is the death rate of virions due to the immune system. All parameters are non-negative.

A second more realistic model also including the elimination of infected cells and virions by the effector cells of the immune system, as well as cell-to-cell transmission is based on [8, 19, 26, 34] and introduced in [5]. In this model which are the effector cells is unspecified. The equations of the model are

$$\begin{aligned}\frac{dx}{dt} &= \lambda - \mu_x x - \beta x v - \beta_{xy} x y \\ \frac{dy}{dt} &= \beta x v + \beta_{xy} x y - \beta_{ey} y e - \mu_y y \\ \frac{dv}{dt} &= B \mu_y y - \beta x v - \beta_{ev} v e - \mu_v v \\ \frac{de}{dt} &= s + \alpha_1 y - \beta_{ye} y e - \beta_{ve} v e - \mu_e e.\end{aligned}\tag{2}$$

Here, x, y, v , and e are the concentrations of susceptible cells, infected cells, virions, and effector cells, respectively. λ is the recruitment rate of new susceptible cells, μ_x is their death rate, β is the infection rate of susceptible cells by free floating virions, μ_y is the killing rate of infected cells by the virus, B is the number of virions produced per infected cell, and μ_v is the death rate of virions. Susceptible cells can also be infected through direct contact with an infected cell at a rate β_{xy} ; effector cells eliminate infected cells at a rate β_{ey} ; free virus particles are introduced into susceptible cells at a rate β ; and effector cells are recruited at a constant rate s and are also recruited depending on the number of infected cells y at a rate α_1 . In addition, effector cells eliminate infected cells and virus particles at rates β_{ye} and β_{ve} , respectively. Finally, effector cells die naturally at a rate μ_e . All parameters are non-negative.

But it is not realistic to consider only one cell type from the immune system since they have different jobs and act at different times. The elimination of the free virions is done by both the innate and adaptive immune systems while the killing of the infected cells is done mostly by the adaptive system. The innate system acts first and then recruits the adaptive system. Herbert in [16] lists neutrophils and natural killer cells as the most important innate cells in fighting virus infections while natural killer T cells are possibly relevant. For the adaptive immune system he has B or plasma cells, cytotoxic T lymphocytes CD4+, and cytotoxic T lymphocytes CD8+. He

mentions several signaling cytokines and factors but only the antibodies as killing agents. He just briefly mentions macrophages but other authors state that they play very important roles [20, 20, 30]. Therefore the variables for the third model are: x susceptible cells, y infected cells, v virions, N neutrophils, m macrophages, K natural killer cells, C_4 CD4+ cells, P plasma cells, C_8 CD8+ cells, a antibodies.

$$\begin{aligned}
\frac{dx}{dt} &= \lambda - \mu_x x - \beta x v - \beta_{xy} x y \\
\frac{dy}{dt} &= \beta x v + \beta_{xy} x y - \beta_{my} m y - \beta_{Ky} K y - \beta_{C_8 y} C_8 y - \mu_y y \\
\frac{dv}{dt} &= B \mu_y y - \beta x v - \beta_{av} a v - \beta_{Nv} N v - \mu_v v \\
\frac{dN}{dt} &= k_N N v - \mu_N N \\
\frac{dm}{dt} &= k_m (m_0 - m) + k_{mv} m v - \beta_{my} m y \\
\frac{dK}{dt} &= k_K m v - \mu_K K \\
\frac{C_4}{dt} &= k_{C_4} K - \mu_{C_4} C_4 \\
\frac{dP}{dt} &= k_P C_4 - \mu_P P \\
\frac{C_8}{dt} &= k_{C_8} - \mu_{C_8} C_8 \\
\frac{da}{dt} &= k_a (a_0 - a) + k_{aP} P - \beta_{va} a v.
\end{aligned} \tag{3}$$

The parameters not include in the previous model are: β_{Ky} the elimination rate of infected cells by neutrophils; $\beta_{C_8 y}$ the elimination rate of infected cells by CD8+ cells; β_{av} the elimination rate of virions by antibodies; β_{Nv} the elimination rate of virions by natural killer cells; k_N the recruitment rate of neutrophils due to the presence of the virus; μ_N the death rate of neutrophils; k_m the natural replenishing rate of macrophages; m_0 the steady state of macrophages in the absence of an infection; k_{mv} the recruitment rate of macrophages due to the virus; β_{my} the elimination of macrophages due to engulfing an infected cell; k_K the activation rate of natural killer cells; μ_K the deactivation rate of natural killer cells; k_{C_4} the activation rate of CD4+ cells; μ_{C_4} the deactivation rate of CD4+ cells; k_P the activation rate of plasma cells; μ_P the deactivation rate of plasma cells; k_{C_8} the activation rate of CD8+ cells; μ_{C_8} the deactivation rate of CD8+ cells; k_a the production rate of antibodies in absence of an infection; a_0 the steady state of the antibodies in absence of an infection; k_{Pa} the production rate of antibodies by activated plasma cells; β_{va} the elimination rate of antibodies after reacting with a virus. All the new parameters are non-negative.

There are some mathematical models that consider different types of immune cells [2, 14, 6] but none has the same immune cells that we do.

There are several different types of antiviral treatments: some block receptors so the virions cannot get into cells, and some lower the viral load by reducing the number of virus copies from an infected cell [9, 10]. Vaccines boost the immune system. They are usually administered before an infection, but they can sometimes be administered after the infection for some slow developing diseases, for example rabies. So we will also consider them as a treatment. Several authors include treatment in their models [24, 31, 7]. The effect of blocking receptors is done by replacing the infection rate β in the equations by $(1 - \epsilon)\beta$. The effect of reducing the viral load by multiplying the number of virions produced by a single cell B by $(1 - \eta)$. Both ϵ and η are between 0 and

1. To our knowledge the effect of strengthening the immune system has not been introduced. We will model it by multiplying the killing rate of virions and infected cells by effector cells by a factor $(1 - \delta)$, with δ between 0 and 1. We will also consider that there is a delay of several days tt between the start of the infection and the administration and action of the antivirals. For example, for the first type of antiviral the factor is $(1 - H(t - tt)\epsilon)$ with H the Heaviside function.

3 Results

We analyzed the three models. For the first model we repeated the analysis done by others [23, 25, 1] and found the two steady solutions (the disease free DFE and a chronic one), their local stability and also calculated the basic reproductive number R_0 , number of susceptible cells infected by a single infected cell at the beginning of the infection [17]. R_0 is given by

$$R_0 \propto \left(\frac{\text{infection}}{\text{contact}} \right) \times \left(\frac{\text{contact}}{\text{time}} \right) \times \left(\frac{\text{time}}{\text{infection}} \right).$$

There are several methods of calculating R_0 . The most widely used approach is the next generation matrix method [12, 29]. R_0 is easier to compute than the eigenvalues of the linearized system since its calculation only involves the infected compartments, y and v . The most important information a mathematical model of infection can give is whether the infection will persist which is for $R_0 > 1$. For the second model, the equilibrium solutions can still be calculated as are their local stability. But the formulas are complicated. The basic reproductive number has a simpler formula and is easier to interpret. For the third model only the DFE and R_0 can be calculated. For all models the numerical simulations give very useful information. We did simulations without treatment and then with treatment. Since the data available has large variations due to differences in the individuals, errors and difficulties in the measurements, different estimation methods, it is very important to do a global sensitivity analysis. This way the relative influence of each parameter on the solutions can be established.

4 Conclusions

The immune system is very complex. There are many different types of cells and factors that interact with themselves, the virus, and healthy and infected cells. Including more processes makes the model more realistic, but in practice it is necessary to limit their number, both to simplify the model and better understand the most important processes. The fact that the number of parameters and data available is limited also restricts the complexity of the model. In this paper we only considered one factor, antibodies, since they kill virions directly. The other factors and cytokines involved in signaling were assumed to have concentrations proportional to the number of cells that produce them. Also for simplicity we did not include some immune cells such as inactivated macrophages. We did comparisons between three models that have different complexities, and studied the effect of introducing antivirals. We also used global sensitivity analysis to establish the parameters that cause the biggest changes in the number of infected cells. This way directions to better measure them can be developed. But most importantly it can encourage research in how to change these parameters to decrease the damage from the viral infection.

References

- [1] BACCAM, P., BEAUCHEMIN, C., MACKEN, C. A., HAYDEN, F. G., AND PERELSON, A. S. Kinetics of influenza a virus infection in humans. *Journal of virology* 80, 15 (2006), 7590–7599.

- [2] BOCHAROV, G. A., AND ROMANYUKHA, A. A. Mathematical model of antiviral immune response iii. influenza a virus infection. *Journal of Theoretical Biology* 167, 4 (1994), 323–360.
- [3] CANINI, L., AND PERELSON, A. S. Viral Kinetic Modeling: State of the Art. *Journal of pharmacokinetics and pharmacodynamics* 41, 5 (Oct. 2014), 431–443.
- [4] CAO, P., WANG, Z., YAN, A. W. C., MCVERNON, J., XU, J., HEFFERNAN, J. M., KEDZIERKA, K., AND MCCAW, J. M. On the Role of CD8+ T Cells in Determining Recovery Time from Influenza Virus Infection. *Frontiers in Immunology* 7 (Dec. 2016), 611.
- [5] CHEN-CHARPENTIER, B. On population models with delays and dependence on past values. *Axioms* 13, 3 (2024), 206.
- [6] CHIMAL-EGUIA, J. C. Mathematical Model of Antiviral Immune Response against the COVID-19 Virus. *Mathematics* 9, 12 (Jan. 2021), 1356.
- [7] CIUPE, S. M. Modeling the dynamics of hepatitis b infection, immunity, and drug therapy. *Immunological Reviews* 285, 1 (2018), 38–54.
- [8] CIUPE, S. M., RIBEIRO, R. M., NELSON, P. W., DUSHEIKO, G., AND PERELSON, A. S. The role of cells refractory to productive infection in acute hepatitis b viral dynamics. *Proceedings of the National Academy of Sciences* 104, 12 (2007), 5050–5055.
- [9] CLARK, M. A., DOUGLAS, M., AND CHOI, J. *Biology 2e*. Openstax, 2018.
- [10] CLINIC, C. What Do Antivirals Treat? <https://my.clevelandclinic.org/health/treatments/antivirals>, 2025.
- [11] CULSHAW, R. V., RUAN, S., AND WEBB, G. A mathematical model of cell-to-cell spread of hiv-1 that includes a time delay. *Journal of mathematical biology* 46, 5 (2003), 425–444.
- [12] DIEKMANN, O., HEESTERBEEK, J., AND ROBERTS, M. G. The construction of next-generation matrices for compartmental epidemic models. *Journal of the royal society interface* 7, 47 (2010), 873–885.
- [13] DU, S. Q., AND YUAN, W. Mathematical modeling of interaction between innate and adaptive immune responses in covid-19 and implications for viral pathogenesis. *Journal of medical virology* 92, 9 (2020), 1615–1628.
- [14] GÓMEZ, M. C., AND YANG, H. M. Mathematical model of the immune response to dengue virus. *Journal of Applied Mathematics and Computing* 63, 1-2 (June 2020), 455–478.
- [15] HATTAF, K., YOUSFI, N., 1 CENTRE RÉGIONAL DES MÉTIERS DE L’ÉDUCATION ET DE LA FORMATION (CRMEF), 20340 DERB GHALEF, CASABLANCA, MOROCCO, AND 2 LABORATORY OF ANALYSIS, MODELING AND SIMULATION (LAMS), FACULTY OF SCIENCES BEN M’SIK, HASSAN II UNIVERSITY OF CASABLANCA, P.O BOX 7955 SIDI OTHMAN, CASABLANCA, MOROCCO. Dynamics of SARS-CoV-2 infection model with two modes of transmission and immune response. *Mathematical Biosciences and Engineering* 17, 5 (2020), 5326–5340.
- [16] HERBERT, J. A., AND PANAGIOTOU, S. Immune Response to Viruses. In *Encyclopedia of Infection and Immunity*. Elsevier, 2022, pp. 429–444.
- [17] JONES, J. H. Notes on r0. department of anthropological sciences, 2007.

- [18] KAJIWARA, T., AND AND SASAKI, T. Global stability of pathogen-immune dynamics with absorption. *Journal of Biological Dynamics* 4, 3 (May 2010), 258–269.
- [19] KIM, H. Y., KWON, H.-D., JANG, T. S., LIM, J., AND LEE, H.-S. Mathematical modeling of triphasic viral dynamics in patients with hbeag-positive chronic hepatitis b showing response to 24-week clevudine therapy. *PLoS One* 7, 11 (2012), e50377.
- [20] MEISCHEL, T., VILLALON-LETELIER, F., SAUNDERS, P. M., READING, P. C., AND LONDRIGAN, S. L. Influenza A virus interactions with macrophages: Lessons from epithelial cells. *Cellular microbiology* 22, 5 (2020), e13170.
- [21] MURASE, A., SASAKI, T., AND KAJIWARA, T. Stability analysis of pathogen-immune interaction dynamics. *Journal of Mathematical Biology* 51, 3 (Sept. 2005), 247–267.
- [22] NOWAK, M. A., AND BANGHAM, C. R. M. Population Dynamics of Immune Responses to Persistent Viruses. *Science* 272, 5258 (Apr. 1996), 74–79.
- [23] NOWAK, M. A., BONHOEFFER, S., HILL, A. M., BOEHME, R., THOMAS, H. C., AND MCDADE, H. Viral dynamics in hepatitis B virus infection. *Proceedings of the National Academy of Sciences* 93, 9 (Apr. 1996), 4398–4402.
- [24] PERELSON, A. S. Modelling viral and immune system dynamics. *Nature Reviews Immunology* 2, 1 (Jan. 2002), 28–36.
- [25] PERELSON, A. S., AND NELSON, P. W. Mathematical Analysis of HIV-1 Dynamics in Vivo. *SIAM Review* 41, 1 (Jan. 1999), 3–44.
- [26] POURBASHASH, H., PILYUGIN, S. S., DE LEENHEER, P., AND MCCLUSKEY, C. Global analysis of within host virus models with cell-to-cell viral transmission. *Discrete Contin. Dyn. Syst. Ser. B* 19, 10 (2014), 3341–3357.
- [27] SHU, H., WANG, L., AND WATMOUGH, J. Global Stability of a Nonlinear Viral Infection Model with Infinitely Distributed Intracellular Delays and CTL Immune Responses. *SIAM Journal on Applied Mathematics* 73, 3 (Jan. 2013), 1280–1302.
- [28] TAVEIRA, N. Antivirals and Vaccines. *International Journal of Molecular Sciences* 24, 12 (June 2023), 10315.
- [29] VAN DEN DRIESCHE, P. Reproduction numbers of infectious disease models. *Infectious Disease Modelling* 2, 3 (2017), 288–303.
- [30] WANG, X.-L., WANG, X., AND HO, W.-Z. Roles of Macrophages in Viral Infections. *Viruses* 16, 10 (Oct. 2024), 1643.
- [31] WANG, Z., AND ZHAO, X.-Q. A Within-Host Virus Model with Periodic Multidrug Therapy. *Bulletin of Mathematical Biology* 75, 3 (Mar. 2013), 543–563.
- [32] XU, Z., SONG, J., ZHANG, H., WEI, Z., WEI, D., YANG, G., DEMONGEOT, J., AND ZENG, Q. A mathematical model simulating the adaptive immune response in various vaccines and vaccination strategies. *Scientific Reports* 14, 1 (Oct. 2024), 23995.
- [33] YANG, Y., ZOU, L., AND RUAN, S. Global dynamics of a delayed within-host viral infection model with both virus-to-cell and cell-to-cell transmissions. *Mathematical Biosciences* 270 (Dec. 2015), 183–191.
- [34] ZHANG, S., LI, F., AND XU, X. Dynamics and control strategy for a delayed viral infection model. *Journal of Biological Dynamics* 16, 1 (2022), 44–63.